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MATHEMATICS 2270
Quiz 4, Fall semester 2012

Name: KEY

1. (5 pts) Project $(1, 0, 3, 1)$ on to the subspace $V = \langle (1, 0, 1, 0), (-1, 2, 2, -1) \rangle$. (You may use the formula $P = A(A^T A)^{-1} A^T$.)

$$A^T A = \begin{pmatrix} 1 & 0 & 1 & 0 \\ -1 & 2 & 2 & -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 2 \\ 1 & 2 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 10 \end{pmatrix} \quad (A^T A)^{-1} = \frac{1}{19} \begin{pmatrix} 10 & -1 \\ -1 & 2 \end{pmatrix}$$

$$\begin{aligned} P v &= \begin{pmatrix} 1 & -1 \\ 0 & 2 \\ 1 & 2 \\ 0 & -1 \end{pmatrix} \frac{1}{19} \begin{pmatrix} 10 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 & 0 \\ -1 & 2 & 2 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & 2 \\ 1 & 2 \\ 0 & -1 \end{pmatrix} \frac{1}{19} \begin{pmatrix} 10 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 4 \\ 4 \end{pmatrix} \\ &= \begin{pmatrix} 1 & -1 \\ 0 & 2 \\ 1 & 2 \\ 0 & -1 \end{pmatrix} \frac{1}{19} \begin{pmatrix} 3 & 6 \\ 4 & 4 \end{pmatrix} = \frac{1}{19} \begin{pmatrix} 3 & 2 \\ 8 \\ 4 & 4 \\ -4 \end{pmatrix} \end{aligned}$$

2. (5 pts) Define the orthogonal complement $V^\perp$ of a subspace $V \subset \mathbb{R}^n$. Find the orthogonal complement of the subspace $V = \langle (1, 0, 1, 0), (-1, 2, 2, -1) \rangle \subset \mathbb{R}^4$.

$$V^\perp = \{ w \in \mathbb{R}^n \mid w \cdot v = 0 \ \forall v \in V \}$$

$$V^\perp = N \begin{pmatrix} 1 & 0 & 1 & 0 \\ -1 & 2 & 2 & -1 \end{pmatrix} = N \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 3 & -1 \end{pmatrix} = N \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 3/2 & -1/2 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ -3/2 & 1/2 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}^T$$

3. (2 pts) If A is an $m \times n$ matrix of rank r then what are the dimensions of $N(A)$, $C(A)$, $N(A^T)$ and $C(A^T)$. (Note that $C(A)$ stands for the column space and $N(A)$ for the null-space.)

$$\dim N(A) = m - r$$

$$\dim C(A) = r$$

$$\dim N(A^T) = n - r$$

$$\dim C(A^T) = r$$